

JEE MAIN REVISION MASTER: CENTER OF MASS AND COLLISION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 128 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Center Of Mass - Core Application	128 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Center Of Mass - Core Application (Very High Yield — 128 PYQs)

Core Principle: Key problem pattern in Center Of Mass.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Two identical bodies A and B of equal masses have initial velocities $\vec{v}_A(1)=4\hat{i}\text{ m/s}$ and $\vec{v}_B(2)=4\hat{j}\text{ m/s}$ respectively. T..." [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "In a perfectly inelastic collision, two spheres made of the same material with masses 15 kg and 25 kg, moving in opposite directions with s..." [Ans: B]
- [2026 Morning Shift] Q#2 (MCQ): "The position of center of mass of three masses 2kg,3kg and 15 kg placed with respect to mid point (p) of normal bisector, as shown in th..." [Ans: A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

Two identical bodies A and B of equal masses have initial velocities $\vec{v}_1 = 4\hat{i} \text{ m/s}$ and $\vec{v}_2 = 4\hat{j} \text{ m/s}$ respectively. The body A has acceleration $\vec{a}_1 = 6\hat{i} + 6\hat{j} \text{ m/s}^2$ while the acceleration of the other body B is zero. The centre of mass of the two bodies moves in _____ path.

- (A) circular
- (B) parabolic
- (C) straight line **[Correct]**
- (D) elliptical

Correct Answer: C

Solution Analysis: For two bodies of equal mass ($m_1 = m_2 = m$), the velocity of the COM is:

$\vec{v}_{cm} = \frac{m\vec{v}_1 + m\vec{v}_2}{m+m} = \frac{\vec{v}_1 + \vec{v}_2}{2}$ The velocity of body A is $\vec{v}_1 = 4\hat{i}$ and that of body B is $\vec{v}_2 = 4\hat{j}$: So, $\vec{v}_{cm} = \frac{4\hat{i} + 4\hat{j}}{2} = (2\hat{i} + 2\hat{j}) \text{ m/s}$ Similarly, for equal masses the acceleration of center of mass is, $\vec{a}_{cm} = \frac{\vec{a}_1 + \vec{a}_2}{2}$ The acceleration of body A is $\vec{a}_1 = 6\hat{i} + 6\hat{j}$ and that of body B is $\vec{a}_2 = 0$: $\vec{a}_{cm} = \frac{(6\hat{i} + 6\hat{j}) + 0}{2}$